

PrimusCredence

Provable Security Solutions
to Cloud, AI & Crypto

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22+ YEARS OF EXPERIENCE IN PROVABLE SECURITY

PhD, IISc Bangalore

Model checking for information-flow security

Oracle Labs · 2014–2019

Java vulnerability detection

ConsenSys R&D · 2019–2021

Cross-chain protocols, consensus verification

Supra · VP of R&D · 2021–ongoing

BFT design & FV, bridges, DeFi

TECHNIQUES

Static analysis, theorem proving, model checking

TOOLS

Dafny, TLA+, IVy, Lean, Isabelle

Why Now: The Attacker Industrialised

A sampling defence cannot answer an enumerating attack.

4.5×

revenue of AI-assisted
scam operations versus
non-AI equivalents

\$3.52

average compute cost per
LLM-agent exploit attempt
on a real CVE

87%

of a 15-CVE sample
exploited from the public
description alone

TRADITIONAL PEN TEST

10^2 – 10^3 paths, human-selected
2–4 week window, then closed
High marginal cost per path
“We found no way in”

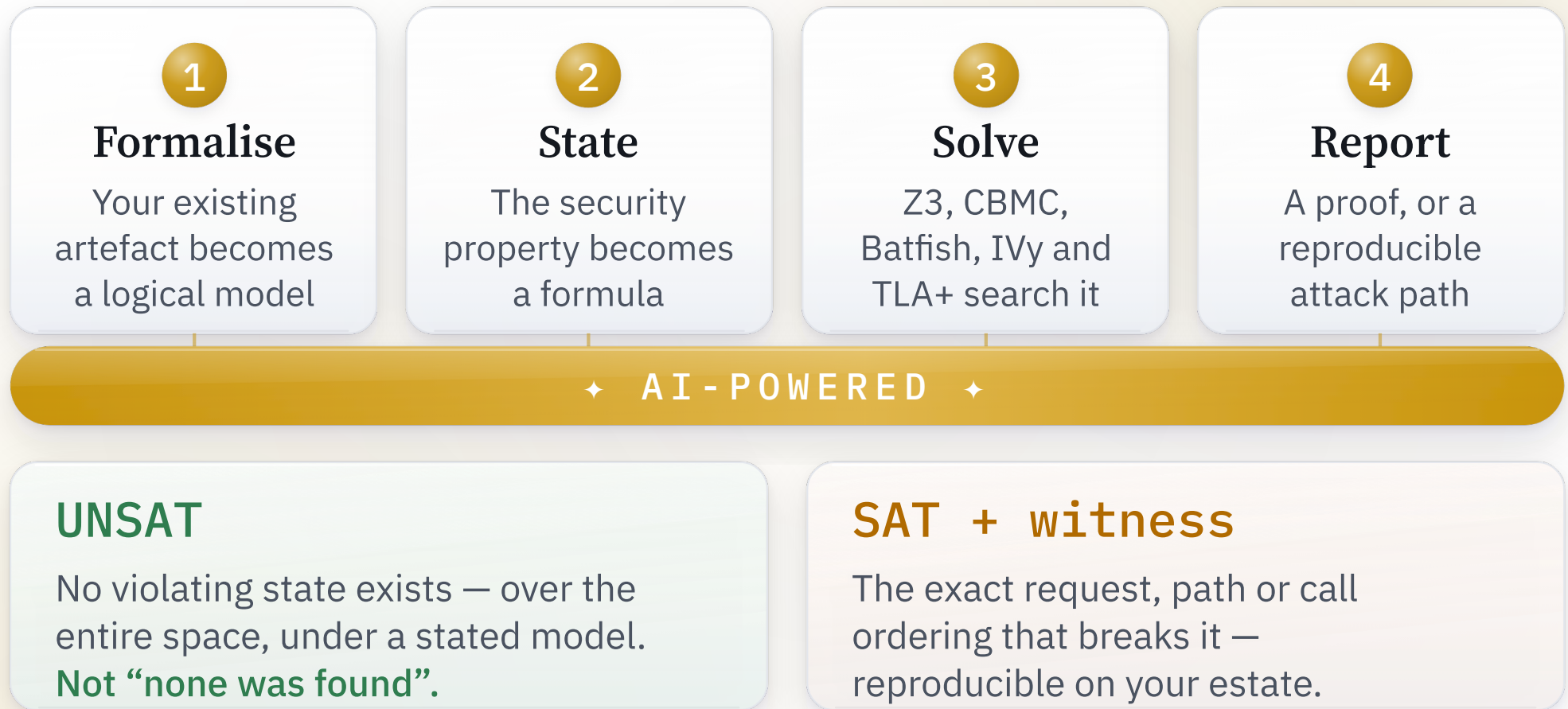
AI-DRIVEN ADVERSARY

10^5 – 10^7 paths, machine-enumerated
Continuous, no window
Near-zero marginal cost
Approaching exhaustive

Proof is the one capability that answers exhaustive search in kind.

Automated Reasoning

Proof establishes that bad things cannot happen.



Industrial practice: AWS (Zelkova, Tiros, Nitro) · Microsoft (Z3) · Meta (Infer) · Airbus (Astrée)

We Add a Layer to Your Stack, NOT Replace It

WHAT YOUR STACK ANSWERS TODAY

Pen test & red team

CSPM / CIEM

ASPM / SAST

Vulnerability management

SOC · EDR · SIEM

“We looked, and within the time available we did not find a way in.”

WHAT WE ADD

“No way it exists — over the whole modelled space.”

A preventive control with evidence behind it, re-run on every change.

WHAT STAYS WITH YOU

Configuration drift, shadow assets and anything outside the model

Insider misuse, and valid credentials used exactly as intended

Detection, response and the rest of the programme you already run

Nine Solutions, Three Families

Cybersecurity

CISO · cloud security
· AppSec

- 1 Cloud entitlement proofs
- 2 Network segmentation proofs
- 3 API-usage conformance
- 4 Firmware & CVE reachability
- 5 Post-quantum migration

AI Security

Head of AI · model
risk

- 6 LLM guardrail verification
- 7 Neurosymbolic assurance

Crypto Security

Protocol lead · CTO ·
VARA

- 8 Smart-contract verification
- 9 Protocol & consensus proofs

Cybersecurity Solutions

- 1 Cloud access-policy and entitlement verification
- 2 Network reachability and segmentation proofs
- 3 API-usage conformance
- 4 Hypervisor, firmware and CVE reachability
- 5 Post-quantum cryptography migration

1. Cloud Access-Policy & Entitlement Verification

Can any principal, in any request context, reach this resource from outside the trust boundary?

THE PROBLEM

IAM is combinatorial: the effective permission composes policies, boundaries, SCPs and request context. CIEM scores the paths it enumerated.

PRECEDENT AWS Zelkova → IAM Access Analyzer
BUYERS Regional CSPs · sovereign hosts

WHAT WE BUILD

- Multi-cloud policies into one logical model
- Privilege-escalation paths across the graph
- Cedar symbolic checks as a CI/CD gate

OUTPUT

Proof of safety over all requests — or the violating request, the escalation path it opens, and its blast radius.

2. Network Reachability & Segmentation Proofs

Does any path exist from an untrusted zone to this subnet — and does east-west segmentation hold?

THE PROBLEM

Probing tests the paths you thought of, at the moment you ran it. Reachability is a property of the configuration, not of the packets sent.

Static: no scanning window, no production traffic — it re-runs on every change.

PRECEDENT AWS Tiros → VPC Reachability Analyzer

BUYERS Sovereign cloud · banks · OT · telcos

OUTPUT

A proof that no path exists — or the path, hop by hop, with the rule that permits it and the owner of that rule.

WHAT WE BUILD

- Batfish models of the hybrid and on-prem estate
- Soufflé and Z3 engines for cloud topologies
- Microsegmentation vs. the declared policy

3. API-Usage Conformance

Does our own code use the platform, credential and crypto SDKs the way their contracts require?

THE PROBLEM

Cloud APIs are protocols, not function calls. Undrained pagination, a logged credential, an unsanitised response — all compile and pass review.

A correct policy invoked through a misused SDK still leaks. Bounded by the library.

PRECEDENT AWS RAPID → CodeGuru · 76% accepted
BUYERS Platform and AppSec teams · ISVs

OUTPUT

The call site, the taint trace, the pattern violated and the corrected call sequence — plus which patterns were checked.

WHAT WE BUILD

- Pattern libraries per SDK, versioned per release
- A conformance gate for library publishers
- PR-gate CI on Soot, CodeQL and Infer

4. Hypervisor, Firmware & CVE Reachability

Can tenant isolation break below the guest — and is this week's hypervisor CVE reachable on our build?

THE PROBLEM

Below the application, security is a memory-safety question, and a flaw there invalidates every control above it. Flashed firmware has no patch path.

Three of four CVEs unreachable turns a fleet emergency into one scheduled patch.

PRECEDENT AWS Nitro · ~330k lines of Isabelle/HOL

BUYERS Sovereign cloud · chip vendors · defence

WHAT WE BUILD

- Firmware and bootloader kits: CBMC and Kani
- Enclave app and attestation-protocol
- CVE triage against your exact build

OUTPUT

Bounded proof of memory safety and isolation, with the bound stated — or the failing trace; per-CVE verdicts ready for VEX.

5. Post-Quantum Cryptography Migration

Where is every cryptographic asset, in what order does it move, and does security hold at each stage?

THE PROBLEM

Harvest-now-decrypt-later opened the window years ago, so detection contributes nothing. NIST settled the algorithms — which is why the work gets mis-scoped.

A programme over quarters with verification as its assurance layer.

PRECEDENT FREAK and Logjam: negotiation failures

BUYERS CSPs · sovereign AI · banks · utilities

OUTPUT

The CBOM and a sequenced migration plan; then, at each stage, a proof the transition preserves the property.

WHAT WE BUILD

- Discovery and a versioned CBOM, gaps listed
- Risk-based sequencing by confidentiality horizon
- Proof of combiner and downgrade states

AI Security Solutions

- 6 Guardrail verification — what an agent may do
- 7 Neurosymbolic assurance — what an agent may say

6. AI and LLM Guardrail Verification

Not “how often does the agent fail?” but “can it do this at all?”

THE PROBLEM

An agent is a non-human identity with delegated privilege and an instruction channel the adversary can write to. A guard model judging a model is detective, not preventive.

*Provable: no rows outside the caller's ACL.
Not provable: no PII ever reaches the user.*

PRECEDENT Bedrock Automated Reasoning

BUYERS Banks · sovereign AI · OT · healthcare

WHAT WE BUILD

- Tool-call constraints over identity and arguments
- Data-access boundaries in the Solution 1 logic
- Composition safety across call sequences

OUTPUT

A result about the mediation layer, not the model: nothing permitted escapes the envelope, and no path bypasses it.

7. Neurosymbolic Assurance

Does this answer follow from the policy we operate under — and if not, what assumption was silently supplied?

THE PROBLEM

Solution 6 bounds what an agent may do; nothing bounds what it may say. Where the output is the product, the failure is entailment: no boundary crossed, liability all the same.

LLM autoformalization drafts the theory; the domain expert ratifies it.

PRECEDENT Bedrock Automated Reasoning checks

BUYERS Banks and insurers · government

OUTPUT

Per answer: the verdict, the derivation or the contradicting rule, and the missing assumption named — an audit trail.

WHAT WE BUILD

- Policy theory development, scope stated
- Entailed / contradicted / unsupported verdicts
- An evaluation harness on the client's corpus

Crypto Security Solutions

- 8 Formal verification of smart contracts
- 9 Formal verification of distributed protocols

8. Formal Verification of Smart Contracts

Does this contract admit a reachable state that breaks its invariants, access control or solvency?

THE PROBLEM

The code is public, the surface permissionless, exploitation atomic and irreversible. Of 2025 DeFi protocol losses, roughly 89% were protocol logic.

“Formally verified” means three different things. The tier decides your risk.

PRECEDENT Certora in CI at Aave, Uniswap, Lido

BUYERS VASPs · DeFi protocols · custody

OUTPUT

Machine-checked invariants — value, solvency, access control, reentrancy, upgrades — or the transaction that breaks them.

WHAT WE BUILD

- Tier 1 fuzzing: Echidna, Medusa, Foundry
- Tier 2 bounded: Certora, Halmos, hevm
- Tier 3 deductive: K/KEVM, Move Prover, Lean

9. Formal Verification of Distributed Protocols

Does this protocol admit an execution that breaks safety under Byzantine faults, asynchrony or partition?

THE PROBLEM

These are design defects: no CVE, no patch, no signature, present from the first commit. One AWS TLA+ finding needed a 35-step sequence.

PRECEDENT Pipelined Moonshot in IVy · FMBC 2024
BUYERS L1/L2 teams · settlement infrastructure

WHAT WE BUILD

- IVy in decidable EPR: proof or counterexample
- Dafny and IronFleet; TLA+ with TLC and TLAPS
- Coq/Rocq for consensus and replication safety

OUTPUT

Machine-checked inductive invariants with the fault, adversary and network assumptions stated — or the breaking execution.

Why Your Clients Will Ask

Four things happened at once in the GCC.

Mandatory replaced voluntary

UAE cyber-resilience obligations now carry penalties of AED 100k–3m.

A regulator named the technique

VARA expects formal verification where applicable for smart contracts.

AI rules arrived before AI assurance

Risk-tiered obligations demand evidence the market cannot yet produce.

Sovereignty is a reachability question

Localisation asks whether regulated data can leave. Solution 2 settles it.

WHERE A FINDING ALSO LANDS

DESC ISR	1 2 3	NESA / UAE IAS	2 4	UAE PDPL	2 3 6
SAMA CSF	1 2 3 6	NCA ECC	1 2 3 4	VARA	8
UAE AI Act	6 7	ISO 27001	1 2 3 5	NIST CSF 2.0	1–5

Regulation routes the finding to a second reader. It is not the reason to commission it.

What a Provider Can Resell

We do not certify — so an audit firm or an MSSP is a channel, not a rival.

Scanning · posture reporting · monitoring



Provable control effectiveness

THE COMMERCIAL MOVE

- Specialist capability inside bids you already win
- You keep the client relationship and the contract

WHY IT DEPLOYS EASILY

- Read-only access, no agent on any host
- No scanning window, no production traffic

Read-only and agentless, so it clears change advisory in one pass.

We produce the technical evidence. Your assessor keeps the opinion.

One Result, Three Registers

First line

Engineering and security operations

The property, its model and assumptions, and either the proof or a reproducible attack path.

Second line

Risk

Likelihood moved off a qualitative guess, with residual risk and the model's limits stated.

Third line

Internal audit and compliance

A clause-mapped control assertion: objective, status, owner and re-verification date.

✓ Design → operating effectiveness · attaches to your GRC records · assessors probe less

The first register is the product. The second and third make it worth more than it cost.

Take Away

**Mathematical rigour to security —
proof alongside testing, across cloud, AI and crypto.**

The shift

Attack path discovery is automated, parallel and cheap. A defence that only samples cannot answer one that enumerates.

The addition

We do not replace testing, monitoring or the SOC. We make one class of controls machine checkable.

The honesty

A proof is relative to a model and a specification — both stated. We produce the evidence; we do not certify.

START HERE — MICRO ASSESSMENT OF SOLUTION 1

One narrow, high-risk slice. Read-only access, no agent deployed, weeks not months — and a counter-example on your own systems, or a proof.

Thank You

*Let us prove one property on your estate —
then judge the method on the result.*

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